Coverage gaps without the reference layers: feature proxies calibrated on public-leaderboard moments

Best Documentation entry · Zindi user **skaiea13** · final selected submissions **gki9K7jy** (public RMSE 0.075383, 16 October 2026) and **6BNL8pZ3** (public RMSE 0.075416, 17 October 2026)

Data: the challenge package (Overture release 2026-08-19.0 extracts, ACS housing units, census tract boundaries, strata tables and the sample submission files) and tract features that another participant derived from the same package and published under the MIT licence (Section 2). No removed reference layer, no copy of one and no outside dataset was used.

Code, outputs and checksums: the full package described in Section 8 is public at https://github.com/HyunsikParker/bias-bounty-coverage-gaps.

1. Summary

The organisers removed the TIGER roads, Microsoft building footprints, facility points and County Business Patterns counts from the package on 25 September. Since then no file in it carries a label or a reference denominator, so a component gap cannot be computed the way the methodology describes. Two things remain: tract features built from the package that track each component, and the public leaderboard, which reports one RMSE per uploaded file.

Our first estimator turned the features into component gaps against estimated denominators and averaged them. Its mean was 0.311, while the public target mean is 0.060, and it scored worse than a constant at its own mean (0.2787 against 0.2663). The estimator behind the selected submissions keeps the features as predictors and takes its scale from the leaderboard:

$$\widehat{y}_i = \bar{y} + \sum_j \beta_j (f_{ij} - m_j).$$

The public mean and variance of the target come from two constant files. The public covariance of each predictor with the target comes from one probe file per predictor, by solving the RMSE of that file for the covariance. The coefficients are $\beta = \Sigma^{-1} c$. The first four joint models scored within 0.00033 of the RMSE we predicted before uploading them; larger models drifted further above their predictions (Table 4), so we stopped adding predictors.

Every step except the leaderboard reads runs from the package with the scripts in `src/`. The reads themselves are listed with their submission IDs (Table 3), and with them the scripts rebuild the selected file gki9K7jy byte for byte and every value of 6BNL8pZ3 (Section 8).

2. Data

File in the package	Use
`reference/<region>/<region>-sample-submission.csv`	Scored GEOIDs and row order. GEOID is read as text, so Maricopa keeps its leading zero.
`reference/<region>/<region>-overture-roads.parquet`, `-overture-buildings.parquet`, `-overture-pois.parquet`	Road length by class and route network, building counts by source, place counts by category (through the derived features below).
`reference/<region>/<region>-census-acs-housing.parquet`	Housing units, used only as a denominator floor.
`strata/<region>/<region>-strata-tract-table.parquet`	`pct_urban`, the urban share of tract population.
`strata/<region>/<region>-census-tracts.parquet`	Tract polygons and land area (through the derived features).

Derived features from another participant. Most predictors are computed from the per-tract counts in `data/features/<region>-features.parquet` of the repository `teddywasserman/overture-coverage-equity`, commit `1c9033ae2373c9c0adbf4f2315bbf6d5485d6fe0` (2 October 2026), MIT licence, copyright 2026 Kapital Experiment. The author linked it in the challenge discussion (thread 35079), and its README states that it is built

only from the challenge's Overture extracts, ACS and strata files. We read 23 of its columns: `region`, `ALAND`, `acs_housing_units`, six road lengths (`road_km_all`, `road_km_named`, `road_km_signed`, `road_km_county_route`, `road_km_signed_lowclass`, `road_km_county_lowclass`), six building counts (`bld_n` and its OpenStreetMap, Microsoft ML, Esri, Google and under-40 m² subsets) and eight place counts (`poi_n`, `poi_fire`, `poi_fire_protection`, `poi_ems`, `poi_schools` and the three "named like a fire station, EMS unit or school but filed in another category" counts). We checked these columns against our own DuckDB build of the same Overture files. Fire, EMS and school place counts agree in all 9,794 tracts. All-place counts agree in 9,734 tracts and building counts in 9,501, with differences of at most three features, which we attribute to how the two builds assign features on tract boundaries. Named-highway length agrees within 10 m in 8,670 tracts (correlation 0.999995; largest difference 2.3 km).

Our own derived file. `analysis/misfiled/classification_deserts.csv` lists 368 tracts (333 of them scored) with no place filed as `fire_department` but at least one place whose name identifies a fire department. It comes from `src/misfiled_stations.py` and `src/misfiled_report.py` (Overture places only) and is the basis of our Best Bias Discovery entry.

Deliberately not used. The strata columns `usfs_BuildingCount_sum`, `usfs_HUCount_sum` and `carbonplan_buildings` count buildings from sources close to the removed footprint reference. Discussion 35302 asks whether they are allowed and had no organiser answer when we built the predictors, so we left them out.

3. Predictors

Each predictor stands in for one part of the published definition (Table 1). They are proxies: none of them recovers a removed denominator, and none needs to, because the calibration in Section 5 only uses how each one co-varies with the target.

Road gap. The definition compares Overture length in the motorway, trunk, primary and secondary classes with TIGER S1100 and S1200 length. Overture's `routes` attribute marks segments that carry an Interstate, US or State route, which correspond closely to the TIGER S1100 and S1200 classes. We estimate the reference length as signed-route length plus county-route length and compute

$$g_i^{road} = 1 - \min\left(1, \frac{L_i^{named}}{L_i^{signed} + L_i^{county}}\right).$$

The component counts as defined ($d_i = 1$) when that estimated reference exceeds 0.1 km. The predictor `road_g1` is $g_i^{road} \cdot d_i/(d_i + 2)$, the road term's share of a three-component mean, and is 0 when the road is undefined. `road_g1_def` is d_i itself. The methodology warns that road gaps are not comparable across regions, so `road_g1_ok` and `road_g1_tx` give Eastern Oklahoma and South-Central Texas their own slopes.

Building gap. The reference is Microsoft's footprint count, and Overture imports many Microsoft footprints, so the composition of a tract's Overture buildings says something about how complete they are. The building predictors are the OpenStreetMap and Esri shares of Overture buildings, the share of footprints under 40 m², and $\log(1 + B_i/\max(HU_i, 1))$, buildings per housing unit.

POI gap, facilities half. The reference counts fire stations, EMS stations and schools. `fire0_rural` equals the rural share, one minus `pct_urban`, when no place is filed as `fire_department`, and 0 otherwise. `fire_evidence` marks tracts with no `fire_department` place that still hold a fire-protection-service place or a fire-department name in another category, and `fire_classdesert` marks the 333 scored classification deserts above. `school0` and `school_evidence` do the same for schools. No EMS predictor entered a fit: its name evidence is non-zero in under 3% of tracts, too sparse for a probe to measure.

POI gap, establishments half. The reference is County Business Patterns establishments. `cbp_lowplaces` is $\mathrm{clip}(1 - r_i/\tilde{r}, 0, 1)$, where r_i is Overture places per housing unit and $\tilde{r} = 0.0861$ its median over tracts with more than 50 housing units.

Region. Indicators for Eastern Oklahoma, Maricopa and Northern California, with South-Central Texas as the base.

Predictor	Component	Construction
`road_g1`	road	estimated road gap times its share of a three-way mean
`road_g1_def`	road	1 if the estimated reference length exceeds 0.1 km
`road_g1_ok`, `road_g1_tx`	road	`road_g1` within Eastern Oklahoma, within South-Central Texas
`bld_osm_share`, `bld_esri_share`	building	share of Overture buildings from OpenStreetMap, from Esri
`bld_small_share`	building	share of footprints under 40 m²
`bld_per_hu_log`	building	log(1 + buildings per housing unit)
`fire0_rural`	facilities	no `fire_department` place, times the rural share
`fire_evidence`	facilities	no `fire_department` place, fire-station evidence elsewhere
`fire_classdesert`	facilities	classification desert (Section 2)
`school0`, `school_evidence`	facilities	no school place; no school place with school-named evidence
`cbp_lowplaces`	establishments	shortfall of places per housing unit against the median
`region_eastern`, `region_maricopa`, `region_northern`	all	region indicators

Table 1. The 17 predictors with a leaderboard read. `src/lbreg_features.py` builds them.

4. Why the first estimator failed

Before any leaderboard read, the first pipeline estimated every removed denominator from housing units, land area and road length, with quantile calibration by urban band, and averaged the defined component gaps. We submitted six of its files on 7 October, each testing one question (Table 2).

File (ID)	Mean	Public RMSE
constant 0.31080482 (6tP83ckS)	0.311	0.266318
three-component composite (YCQGYJzr)	0.311	0.278726
building component only (ctLe5wRK)	0.232	0.324419
denominators raised 25% (Xng1Jyqn)	0.407	0.338412
probabilistic road definition (UcmHP36M)	0.313	0.416334

Table 2. First-pipeline reads. A sixth file without the unscored Eastern Washington rows was rejected.

The composite lost to a constant at the same mean, and raising the denominators made it worse. The target moments (Section 5.1) explain why. The target averages 0.060 with standard deviation 0.089, so most tracts have almost no gap, and a model centred at 0.31 misses nearly every tract by more than any variation it captures. The constant 0.060 alone would score 0.089. The package holds nothing that fixes this scale, so we took it from the leaderboard (Section 5).

5. Calibration from public-leaderboard moments

5.1 Target moments

Write P for the public split (about 30% of the 9,379 scored tracts). For a constant prediction k, $\mathrm{RMSE}^2 = E_P[y^2] - 2k\,\bar{y} + k^2$. An all-zero file scored 0.107246 after the 2 October scorer change, as participant Kimyayd reported in thread 35079, which gives $E_P[y^2] = 0.0115017$. Our constant 0.31080482 scored 0.266318147. Together they give $\bar{y} = 0.059806$ and a standard deviation of 0.089022. The same participant reported 0.059806 for the all-zero file under the scorer before 2 October and read that metric as a mean absolute error. For a non-negative target and an all-zero file, the mean absolute error equals the mean, so the two values agree. The scripts carry $\bar{y} = 0.059802$, which we derived with the constant rounded to 0.3108. The 0.000004 difference shifts every prediction by that amount and changes no reported RMSE.

5.2 Probes

A probe for predictor f predicts $\bar{y} + a(f_i - m)$ in the four scored regions, where m is the mean of f over the scored tracts, and $\bar{y}$ in Eastern Washington. The file format requires the Eastern Washington rows, but they are not scored. Expanding the squared error gives

$$\mathrm{RMSE}^2 = \mathrm{Var}_P(y) - 2a\,\mathrm{Cov}_P(y, f) + a^2 E_P\left[(f - m)^2\right],$$

which we solve for $\mathrm{Cov}_P(y, f)$. Since the scored-tract value of $E[(f - m)^2]$ stands in for its public value, the read assumes that the public split is a random subset. We chose each a so that predictions stay in [0, 1] and the RMSE moves by far more than the nine displayed decimals. `src/lbreg_probe.py` writes a probe file and converts a read into a covariance, a correlation and the best RMSE that predictor alone could reach.

ID	Predictor	a	Public RMSE	Cov(y, f)	Corr	Best alone
4Yru3kU9	`road_g1`	1	0.088372	+0.001222	+0.285	0.085345
cF851kiC	`fire_evidence`	0.1	0.089489	+0.002774	+0.123	0.088345
pEvURwmn	`fire0_rural`	0.1	0.086352	+0.005664	+0.247	0.086269
4WMt2RbL	`region_eastern`	0.05	0.085751	+0.008496	+0.287	0.085293
MHgJAeDX	`region_maricopa`	0.05	0.092588	-0.002946	-0.088	0.088679
jCsD82GY	`region_northern`	0.05	0.089567	+0.000508	+0.023	0.089000
tyzLSRnx	`cbp_lowplaces`	0.05	0.090402	-0.000703	-0.030	0.088986
5W8b8P4f	`school0`	0.05	0.089951	+0.002581	+0.070	0.088804
iMHPKR4w	`road_g1_def`	0.05	0.086558	+0.009668	+0.235	0.086531
jiNeSpzk	`road_g1_ok`	1	0.085544	+0.000819	+0.287	0.085291
gZtnGknB	`road_g1_tx`	1	0.092420	+0.000372	+0.113	0.088452
LJn8pYVm	`bld_osm_share`	0.1	0.107297	-0.010479	-0.305	0.084789
N2eHmQEE	`fire_classdesert`	0.1	0.087474	+0.003081	+0.187	0.087454
ykjXfBWm	`bld_per_hu_log`	0.05	0.087883	+0.007638	+0.181	0.087555
y8wcxgqQ	`bld_esri_share`	0.17	0.092582	-0.000163	-0.013	0.089018
KZf55RXS	`bld_small_share`	0.3	0.090287	+0.000564	+0.080	0.088740
kBNDdBt4	`school_evidence`	0.1	0.091912	+0.000354	+0.016	0.089013
yhmsidry	`fire0_any`	0.06	0.094578	-0.002284	-0.056	0.088883
871Ypj56	`road_g0_def`	0.05	0.085097	+0.012951	+0.294	0.085084
JXywjudP	`fire0_logarea`	0.02	0.088213	+0.027013	+0.198	0.087257
bNSxfWJe	`fire0_noroad`	0.06	0.097896	-0.008129	-0.210	0.087046
USMyMrTA	`ems_evidence`	0.17	0.091696	+0.000751	+0.053	0.088901

ID	Predictor	a	Public RMSE	Cov(y, f)	Corr	Best alone
wRTfDyWv	`bld_google_share`	0.35	0.092376	+0.000218	+0.031	0.088982
rcTdTnXi	`fire0_rural_noroad`	0.45	0.093310	-0.000092	-0.018	0.089011
5McWTXq7	`road_lowclass_share`	0.9	0.090515	+0.000146	+0.064	0.088842
pzB6kHpi	`road_g05_def`	0.058	0.086645	+0.009826	+0.238	0.086460
V9xHm5JM	`road_county_low_share`	0.18	0.085701	+0.003630	+0.272	0.085658

Table 3. Single-predictor probes, 11 to 17 October. The constant at the mean scores 0.089022. `probes/lbreg_probes.json` holds the a and RMSE columns that `src/lbreg_fit.py` reads.

5.3 Joint estimator

Σ is the covariance matrix of the chosen predictors over all scored tracts, so it uses every tract rather than the public 30% only. c is the vector of probed covariances. Then $\beta = \Sigma^{-1}c$, and before clipping the expected public RMSE is $\sqrt{\mathrm{Var}_P(y) - c^\top \Sigma^{-1} c}$. Predictions are clipped to [0, 1]: one tract falls below zero in the 13-predictor model, 28 in the 17-predictor, 64 in the 21-predictor and 82 in the 25-predictor model, and none exceeds one.

ID	Model	Expected	Public RMSE
8R4BSBA6	5 predictors (`road_g1`, `fire_evidence`, `fire0_rural`, two regions)	0.081292	0.081498
7xysJ25i	9 predictors (adds `region_northern`, `cbp_lowplaces`, `school0`, `road_g1_def`)	0.079606	0.079857
FNBceCuK	13 predictors (adds `road_g1_ok`, `road_g1_tx`, `bld_osm_share`, `fire_classdesert`)	0.078400	0.078660
knE1GJ3o	17 predictors (adds `bld_per_hu_log`, `bld_esri_share`, `bld_small_share`, `school_evidence`)	0.075837	0.076163
gki9K7jy	21 predictors (adds `fire0_any`, `road_g0_def`, `fire0_logarea`, `fire0_noroad`)	0.074989	0.075383
6BNL8pZ3	25 predictors (adds `ems_evidence`, `bld_google_share`, `fire0_rural_noroad`, `road_lowclass_share`)	0.074839	0.075416
1Vtrct6A	27 predictors (adds `road_g05_def`, `road_county_low_share`)	0.074725	0.075466

Table 4. Joint models. From knE1GJ3o on, every file applies the no-Overture rule of Section 6, and its expectation assumes one public water tract with target 1 among about 2,814 public tracts (without the rule the 17-, 21-, 25- and 27-predictor fits expect 0.077938, 0.077127, 0.076980 and 0.076868). The first four reads land 0.0002 to 0.0003 above their predictions; the gap then grows to 0.0004, 0.0006 and 0.0007, and the 25- and 27-predictor files read worse than the 21-predictor one. gki9K7jy and 6BNL8pZ3 are the two files selected for private scoring.

The coefficients of the 13-predictor file, FNBceCuK, are in `outputs/joint13.json`. The largest are on `road_g1` (+0.356), `road_g1_tx` (−0.222), `road_g1_ok` (+0.071), `fire_classdesert` (+0.056), `region_eastern` (+0.045) and `bld_osm_share` (−0.032). The road shortfall carries the most signal, with less weight in South-Central Texas and more in Eastern Oklahoma. A classification desert raises the prediction, as the facility definition implies: a station filed outside `fire_department` counts as missing. A high OpenStreetMap share lowers it, which suggests that where OpenStreetMap supplies the buildings, Overture holds at least as many as the Microsoft reference.

The selected files keep this pattern. In gki9K7jy (`outputs/joint21.json`) the largest coefficients are on `road_g1` (+0.412), `road_g1_tx` (−0.294), `bld_esri_share` (−0.065), `fire_classdesert` (+0.054), `bld_small_share` (+0.051) and `region_eastern` (+0.049). 6BNL8pZ3 (`outputs/joint25.json`) has the same leading terms, with `road_g1` at +0.388 and `road_g1_tx` at −0.280.

6. Edge cases

- **GEOID as text.** Every script reads GEOID as a string, and every submission keeps the sample files' GEOIDs and row order, so Maricopa's `04…` tracts keep their leading zero and every join succeeds.

- **Eastern Washington.** The scorer rejects a file without its 415 rows, but they are not scored. They are set to $\bar{y}$ and excluded from every mean, covariance and fit.
- **Tracts with no Overture data.** Two scored tracts, 48039990000 and 48057990000 (Texas water tracts), hold no Overture road, building or place. A tract in the authoritative sample has at least one defined component, and with a zero Overture numerator any defined component has a gap of 1, so their score is 1. FNBceCuK predicts 0.0527 for both, because the linear model sees only all-zero features. `src/edge_no_overture.py` applies the rule from the 15 October files on, including both selected files. The first, knE1GJ3o (17 predictors), read 0.076163 on 15 October: one public tract with target 1 gives 0.075837 plus the offset of Table 4 (none would give 0.077938, both 0.073677).
- **Zero-population and zero-housing tracts.** The 26 scored tracts with no population are exactly the tracts with no `pct_urban`. The code treats a missing urban share as fully urban, which sets `fire0_rural` to 0 rather than guessing a rural share. 80 tracts have no housing units and 101 have 50 or fewer. Every per-housing-unit quantity divides by $\max(HU_i, 1)$, and `cbp_lowplaces` is set to 0 at 50 housing units or fewer, where a single shop moves places per housing unit a long way.
- **Undefined road component.** 2,897 scored tracts (30.9%) have no estimated reference road: 64.3% in Maricopa, 31.3% in Northern California, 23.7% in Eastern Oklahoma and 23.4% in South-Central Texas. The organisers report that at least one component is undefined in 55%, 37%, 21% and 28% of tracts in the same regions. That is not the same quantity, but Maricopa stands far above the others in both. For such tracts `road_g1` is 0 and `road_g1_def` lets the fit move their level, which stands in for the varying divisor of the reference mean.
- **Data-sparse tracts.** 70.7% of scored tracts have no `fire_department` place and 21.6% no school place. A zero count cannot tell a tract without a station from a tract whose station is missing, so the facility predictors weight zero counts by rurality (`fire0_rural`) or require evidence that a station exists under another category (`fire_evidence`, `fire_classdesert`, `school_evidence`).

7. Alternative weightings and why

- **Equal weights on proxy components** (Section 4) put the mean at 0.31 and lost to a constant. With the reference denominators gone, nothing in the package can set the scale of the components.
- **Single predictors** (Table 3) reach at best 0.0848 (`bld_osm_share`). The predictors carry partly separate information, so the 13-predictor model reaches 0.0787.
- **Region-specific road slopes.** The road predictor restricted to Eastern Oklahoma correlates with the target at 0.287 and the one restricted to South-Central Texas at 0.113, in line with the organisers' note that the TIGER-to-Overture length ratio differs by region (0.71 to 1.59).
- **No shrinkage; selection stopped at 25 predictors.** The fits estimate 13 to 27 coefficients from covariances measured on about 2,800 public tracts. The optimism of an in-sample fit is of order p/n, under 1% of the variance here, so we did not shrink the coefficients. Table 4 shows the cost growing with p: the read sits 0.0003 above prediction at 17 predictors and 0.0007 above it at 27, and the 25- and 27-predictor files read worse than the 21-predictor one. We selected the 21-predictor file, our best public read, and the 25-predictor file, and stopped probing.

8. Reproducing the submissions

The package (linked at the top) runs on Python 3.13 with pandas, NumPy and pyarrow. DuckDB's command-line tool with its spatial extension is needed only to rebuild the classification-desert list, which is included. Nothing in the pipeline is random.

1. Copy the challenge package so that `data/reference/<region>/` and `data/strata/<region>/` sit next to `src/`, for example with `aws s3 sync s3://us-west-2.opendata.source.coop/humane-intelligence/bias-bounty-mapping-equity-challenge/reference/ data/reference/ --no-sign-request` and the same command for `strata/`.

2. Put the five `data/features/<region>-features.parquet` files of `teddywasserman/overture-coverage-equity` at commit `1c9033ae2373c9c0adbf4f2315bbf6d5485d6fe0` in `third_party/overture-coverage-equity/data/features/`. `SHA256SUMS` lists their hashes.
3. Optional: `python src/misfiled_stations.py` then `python src/misfiled_report.py` rebuild `analysis/misfiled/classification_deserts.csv`.
4. `python src/lbreg_features.py` writes `work/lbreg_features.parquet` in about two seconds.
5. `python src/lbreg_fit.py outputs/joint13.csv road_g1 fire_evidence fire0_rural region_eastern region_maricopa region_northern cbp_lowplaces school0 road_g1_def road_g1_ok road_g1_tx bld_osm_share fire_classdesert` rebuilds the file submitted as FNBceCuK, SHA-256 `1969f09d1f715c173dde2456cc5ffaa9a2d7582e729704183e982cabc12aabaf`. The same command for `outputs/joint17.csv` with the 17 predictors of `outputs/joint17.json`, then `python src/edge_no_overture.py outputs/joint17.csv outputs/joint17_w1.csv`, rebuilds knE1GJ3o.
6. The two selected files are built the same way. `python src/lbreg_fit.py outputs/joint21.csv` followed by the 21 predictors listed in `outputs/joint21.json`, then `python src/edge_no_overture.py outputs/joint21.csv outputs/joint21_w1.csv`, rebuilds gki9K7jy byte for byte, SHA-256 `604eea81067bcebfc67df3b864fc3d7a7ec3692b4fc28de148861b1c8596ba66`. The same two commands with the 25 predictors of `outputs/joint25.json` give every value of 6BNL8pZ3. The uploaded file differs only in format: it has CRLF line endings and writes the two water tracts as `1` rather than `1.0` (uploaded SHA-256 `254eabbde873bb79df9579614e0aa371e036b78ef5186be1ad64ffc9dbe2afbe`, rebuilt `4c22b3a285021fcf0894dd6cf3b1a7b10d80c0a8fd7c40fa4938f9494e0bf25a`).
7. A new probe is written with `python src/lbreg_probe.py write <predictor> <a> <file.csv>` and read with `python src/lbreg_probe.py read <predictor> <a> <public RMSE>`.

We ran steps 2, 4, 5 and 6 in a fresh folder on 2 November: the rebuilt FNBceCuK, knE1GJ3o and gki9K7jy files are byte-identical to the uploaded ones, and the rebuilt 6BNL8pZ3 file matches the uploaded values exactly.

9. Limitations

- The coefficients rest on the public 30%. If the split is not random, or the private tracts relate to the predictors differently, the private score will be worse than the public one by more than the usual noise. On the private leaderboard shown on 1 November, while the top solutions were still under code review, gki9K7jy read 0.078014 and 6BNL8pZ3 0.078044, both about 0.0026 above their public reads. That is more than the roughly 0.0004 the p/n optimism of 21 coefficients would explain, which points to differences between the public and private tracts or to more noise in the probed covariances than we assumed.
- Each covariance comes from one read rounded to nine decimals, and $E_P[(f-m)^2]$ is replaced by its value over all scored tracts. Table 4 bounds the combined error at about 0.0003 in RMSE for the models we submitted.
- The predictors rank tracts; they do not estimate any removed reference count, and the model does not produce component scores.
- Most predictors come from another participant's feature files. We verified the counts we use against our own build (Section 2), but any error common to both builds would pass that check.
- The method uses the public leaderboard as a measuring instrument: each read yields one covariance. We list every read so that a reader can judge that choice and reproduce the coefficients.